

Training dynamics of overparameterized neural networks

Overview of results

Shreyas Kalvankar

May 22, 2026

TU Delft

Problem setup

We study regression on the unit circle

$$S^1 \simeq [0, 2\pi), \quad x(\gamma) = (\cos \gamma, \sin \gamma).$$

The target is a finite Fourier mixture:

$$f^*(\gamma) = a_0 + \sum_{k=1}^K (a_k \cos(k\gamma) + b_k \sin(k\gamma)).$$

Question: which Fourier components are learned quickly, and what changes at finite sample size and finite width?

Training dynamics

Consider a one-hidden-layer ReLU network

$$f_{\theta}(x) = \frac{1}{\sqrt{m}} \sum_{\alpha=1}^m v_{\alpha} \sigma(w_{\alpha}^{\top} x + b_{\alpha}), \quad x \in S^1,$$

where $\theta = \{(v_{\alpha}, w_{\alpha}, b_{\alpha})\}_{\alpha=1}^m$, initialized via $v_{\alpha}, w_{\alpha} \sim \mathcal{N}(0, 1)$ and $b_{\alpha} = 0$ denotes the set of trainable parameters.

We train with squared loss in continuous time:

$$\mathcal{L}(\theta) = \frac{1}{2} \|f_{\theta} - f^*\|_{L^2(S^1)}^2.$$

Define the residual

$$r_t(x) = f_{\theta_t}(x) - f^*(x).$$

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From parameter flow to the tangent kernel

Gradient flow in parameter space is

$$\frac{d}{dt}\theta_t = -\nabla_{\theta}\mathcal{L}(\theta_t).$$

Since

$$\mathcal{L}(\theta) = \frac{1}{2} \int_{S^1} (f_{\theta}(y) - f^*(y))^2 d\mu(y),$$

we have

$$\nabla_{\theta}\mathcal{L}(\theta_t) = \int_{S^1} r_t(y) \nabla_{\theta}f_{\theta_t}(y) d\mu(y).$$

Using the chain rule, for fixed x ,

$$\frac{d}{dt}f_{\theta_t}(x) = \langle \nabla_{\theta}f_{\theta_t}(x), \dot{\theta}_t \rangle = \langle \nabla_{\theta}f_{\theta_t}(x), -\nabla_{\theta}\mathcal{L}(\theta_t) \rangle.$$

Substituting the gradient flow gives

$$\frac{d}{dt}f_{\theta_t}(x) = - \int_{S^1} \underbrace{\langle \nabla_{\theta}f_{\theta_t}(x), \nabla_{\theta}f_{\theta_t}(y) \rangle}_{\Theta_t(x,y)} r_t(y) d\mu(y).$$

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Residual dynamics in function space

The neural tangent kernel is

$$\Theta_t(x, y) = \langle \nabla_{\theta} f_{\theta_t}(x), \nabla_{\theta} f_{\theta_t}(y) \rangle.$$

Define an integral operator

$$(T_t g)(x) = \int_{S^1} \Theta_t(x, y) g(y) d\mu(y).$$

Thus,

$$\frac{d}{dt} r_t(x) = \frac{d}{dt} f_{\theta_t}(x) = - \int_{S^1} \Theta_t(x, y) r_t(y) d\mu(y) = -T_t r_t$$

If $T_t \equiv T$ is fixed, then

$$r_t = e^{-tT} r_0.$$

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Infinite-width NTK reference

At finite width, the tangent kernel is an empirical average over neurons:

$$\Theta_t^{(m)}(x, y) = \frac{1}{m} \langle \nabla_{\theta} f_{\theta}(x), \nabla_{\theta} f_{\theta}(y) \rangle$$

Taking limit $m \rightarrow \infty$,

$$\Theta_t^{(m)}(x, y) \longrightarrow \Theta_t^{(\infty)}(x, y) = \mathbb{E}_{\theta_0} [\langle \nabla_{\theta} f_{\theta_0}(x), \nabla_{\theta} f_{\theta_0}(y) \rangle].$$

and the limit stays fixed during training:

$$\Theta_t^{(\infty)} \approx \Theta_0^{(\infty)} \approx \Theta_{\infty}.$$

On S^1 , write $x = x(\gamma)$, $y = x(\psi)$, and

$$\delta = |\gamma - \psi|, \quad \delta \in [0, \pi].$$

Then

$$\Theta_{\infty}(x(\gamma), x(\psi)) = \Theta(\delta) = \frac{1}{2\pi} \left(\sin \delta + 2(\pi - \delta) \cos \delta + (\pi - \delta) \right).$$

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Spectral bias in the continuum reference model

Since $\Theta_\infty(x(\gamma), x(\psi)) = \Theta(\gamma - \psi)$, the operator is a convolution:

$$(T_\infty g)(\gamma) = \int_0^{2\pi} \Theta(\gamma - \psi) g(\psi) \frac{d\psi}{2\pi}.$$

Therefore the Fourier basis diagonalizes T_∞ :

$$T_\infty \phi_p = \lambda_p \phi_p.$$

Write the residual as

$$r_t = \sum_p \alpha_p(t) \phi_p.$$

Then

$$\frac{d}{dt} \alpha_p(t) = -\lambda_p \alpha_p(t), \quad \alpha_p(t) = e^{-\lambda_p t} \alpha_p(0).$$

Spectral bias: modes with larger λ_p decay faster.

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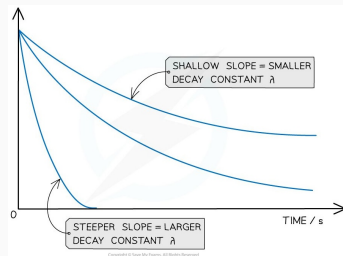
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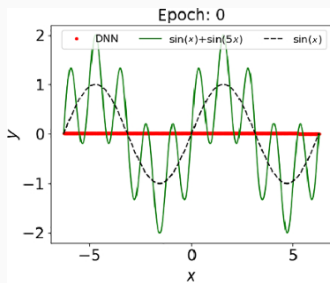
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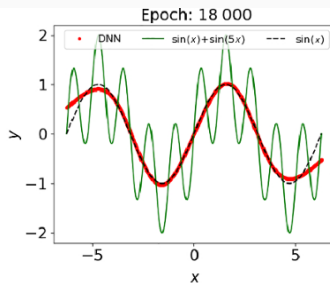
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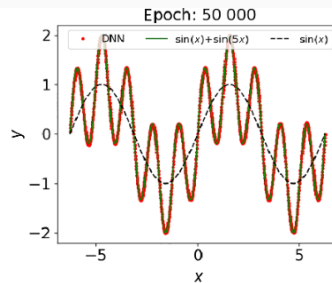
Spectral bias



(a)



(b)



(c)

Image source: *Overview of Spectral Bias in Deep Learning*, Vol. 7

Main research question

Ideal reference:

$$\frac{d}{dt}r_t = -T_\infty r_t, \quad T_\infty \phi_p = \lambda_p \phi_p.$$

Question:

How much of this Fourier-diagonal picture survives when we move away from the idealized setting?

1. finite samples n
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Finite samples

Sample points:

$$\gamma_1, \dots, \gamma_n \sim \text{Unif}[0, 2\pi).$$

Restrict to

$$\mathcal{H}_K = \text{span}\{1, \sqrt{2} \cos \gamma, \sqrt{2} \sin \gamma, \dots, \sqrt{2} \cos K\gamma, \sqrt{2} \sin K\gamma\}.$$

Collect sampled Fourier features in

$$\Phi_{i,p} = \phi_p(\gamma_i), \quad \Phi \in \mathbb{R}^{n \times (2K+1)}.$$

Let $A \in \mathbb{R}^{n \times n}$ be the normalized sampled kernel matrix:

$$A_{ij} = \frac{1}{n} \Theta(\gamma_i - \gamma_j).$$

We wish to find how do these behave,

$$G = \frac{1}{n} \Phi^\top \Phi, \quad H = \frac{1}{n} \Phi^\top A \Phi.$$

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We wish to find how do these behave,

$$G = \frac{1}{n} \Phi^\top \Phi, \quad H = \frac{1}{n} \Phi^\top A \Phi.$$

With high probability,

$$G \approx I, \quad H \approx \Lambda^{(n)}.$$

Meaning: finite sampling perturbs the Fourier-diagonal picture, but on fixed H_K , Fourier modes are almost orthogonal and decay rates are still governed by the continuum eigenvalues.

Proof idea: entrywise concentration

Gram matrix G

$$G_{pq} = \frac{1}{n} \sum_{i=1}^n \phi_p(\gamma_i) \phi_q(\gamma_i).$$

$$\mathbb{E}[G_{pq}] = \delta_{pq}.$$

Entry = average of independent bounded variables.

$\Rightarrow$ Bernstein

Apply the inequality entrywise, then union bound over all p, q :

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Compressed operator H

$$H_{pq} = \frac{1}{n^2} \sum_{i,j=1}^n \phi_p(\gamma_i) \Theta(\gamma_i - \gamma_j) \phi_q(\gamma_j).$$

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Changing one sample changes only $O(n)$ terms.

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Consequence: the sampled operator remains approximately Fourier-diagonal on the subspace. 11/25

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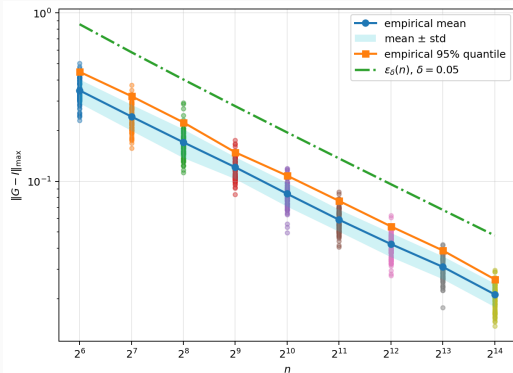
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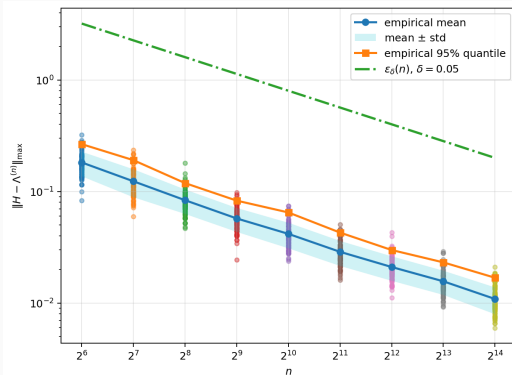
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Empirical finite-sample concentration

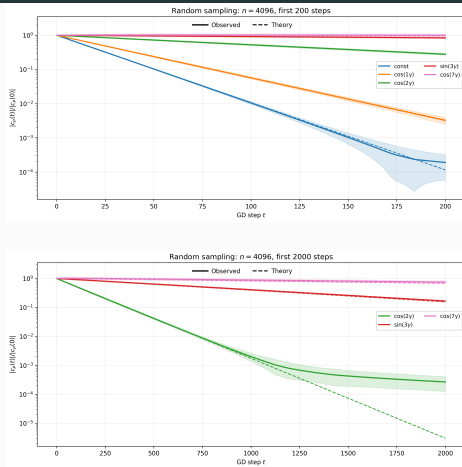


Gram matrix concentration: $\|G - I\|_{\max}$



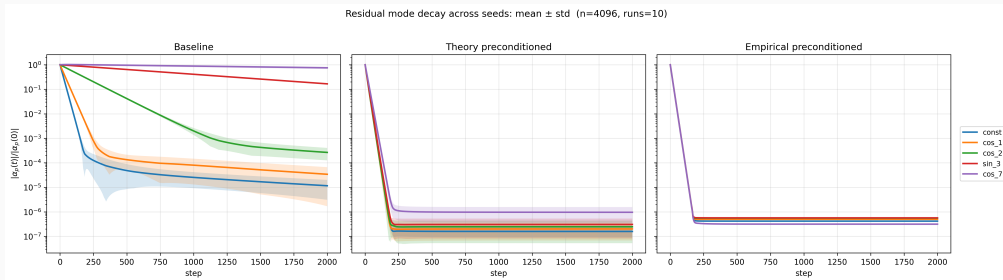
Compressed operator concentration: $\|H - \Lambda^{(n)}\|_{\max}$

Finite-sample dynamics: observed vs diagonal prediction



Top: early dynamics for all retained components. Bottom: later dynamics for higher-frequency components.

Using the spectrum for preconditioning

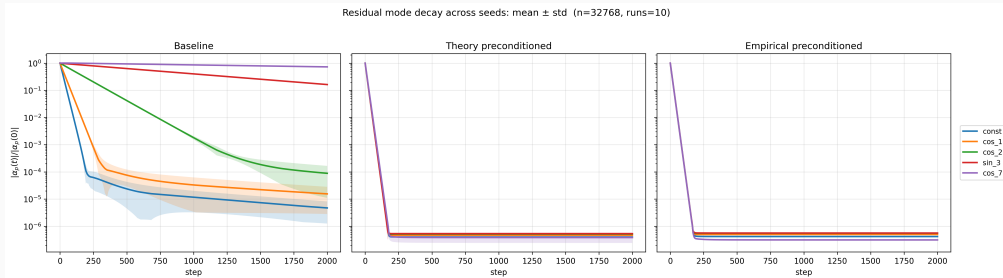


Based on the sampled spectral picture, we can ask whether the dynamics can be preconditioned so that all modes in $\mathcal{H}_K$ decay on a comparable time scale.

Theory preconditioner: uses only the predicted eigenvalues $\lambda_p^{(n)}$.

Empirical preconditioner: uses the observed sampled operator.

Increasing sample size improves theory-based preconditioning



Keeping the same setup but increasing the sample size improves the spectral estimate, so the theory-based preconditioner gets closer to the empirically preconditioned one.

Takeaway: as n increases, the finite-sample eigenvalue prediction becomes more accurate, and preconditioning based only on $\Lambda^{(n)}$ becomes nearly as effective as using the empirical operator.

Finite width: frozen operator at initialization

Now keep the input space continuous, but use finite width m .

$$T_0^{(m)} = \frac{1}{m} \sum_{r=1}^m T^{(r)}.$$

Here $T^{(r)}$ is the one-neuron tangent operator at initialization.

Let $P_K : L^2(S^1) \rightarrow \mathcal{H}_K$ be an orthogonal projection. Project to the retained Fourier subspace:

$$B_m^{(K)} = P_K T_0^{(m)} P_K, \quad \Lambda_K = P_K T_\infty P_K.$$

For Fourier modes ϕ_p, ϕ_q ,

$$(B_m^{(K)})_{pq} = \frac{1}{m} \sum_{r=1}^m \xi_{r,pq}, \quad \xi_{r,pq} = \langle \phi_p, T^{(r)} \phi_q \rangle.$$

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Finite width: concentration around the infinite-width block

Since the neurons are i.i.d.,

$$\mathbb{E}[B_m^{(K)}] = \Lambda_K.$$

Each entry is an average of independent one-neuron contributions:

$$(B_m^{(K)})_{pq} - (\Lambda_K)_{pq} = \frac{1}{m} \sum_{r=1}^m \left(\xi_{r,pq} - \mathbb{E} \xi_{r,pq} \right).$$

The variables are not bounded, but Gaussian initialization gives sub-exponential tails.

$\xi_{r,pq}$ sub-exponential

Bernstein plus a union bound gives

$$\|B_m^{(K)} - \Lambda_K\|_{\max} \lesssim \sqrt{\frac{\log d_K}{m}} \quad \text{with high probability.}$$

Finite width: concentration around the infinite-width block

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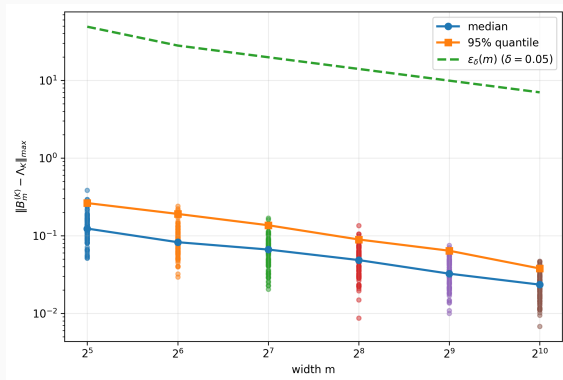
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Empirical finite-width concentration



The bound is conservative because it is proved entrywise, uses a uniform tail bound for all Fourier pairs, and then applies a union bound over the whole $d_K \times d_K$ block. It also controls worst-case high-probability behavior, not the typical median behavior.

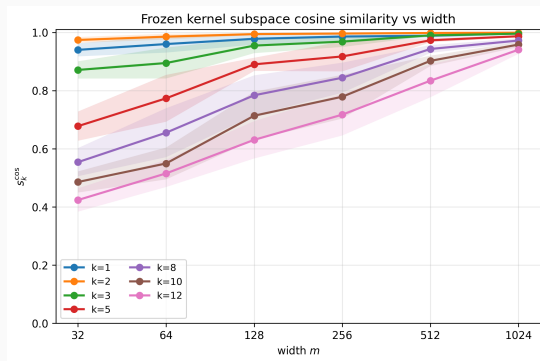
Finite-width eigenspaces vs Fourier planes

For each frequency $k \geq 1$, the relevant continuum eigenspace is the two-dimensional Fourier plane

$$\mathcal{F}_k = \text{span}\{\phi_{k,c}, \phi_{k,s}\}.$$

At finite width, we compare this with the matched empirical eigenspace

$$\hat{\mathcal{F}}_k^{(m)}.$$



Finite-width eigenspaces vs Fourier planes

The two subspaces are both 2-dimensional, so they can be viewed as two planes in a high-dimensional space.

The principal angles

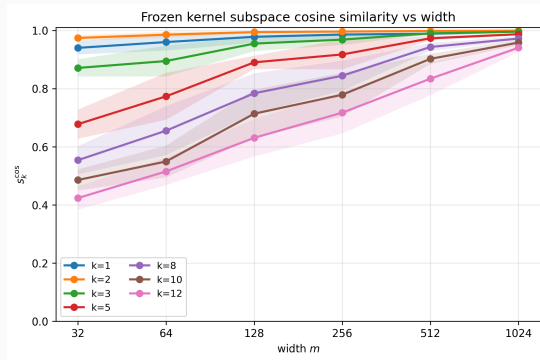
$$0 \leq \theta_{k,1} \leq \theta_{k,2} \leq \frac{\pi}{2}$$

measure how well these two planes align.

We summarize this by the subspace cosine similarity

$$s_k^{\text{cos}} = \left(\frac{1}{2} \sum_{i=1}^2 \cos^2 \theta_{k,i} \right)^{1/2}.$$

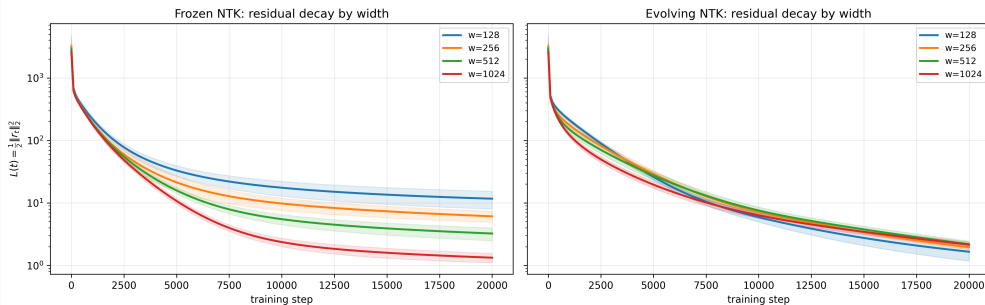
$$s_k^{\text{cos}} = 1 \iff \widehat{\mathcal{F}}_k^{(m)} = \mathcal{F}_k.$$



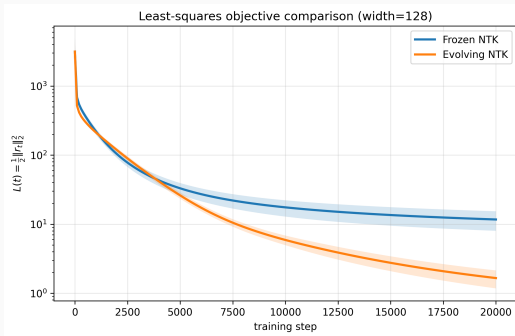
Frozen versus evolving finite-width dynamics

$$\text{Frozen: } \frac{d}{dt} r_t = -T_0^{(m)} r_t \quad \text{Evolving: } \frac{d}{dt} r_t = -T_t^{(m)} r_t$$

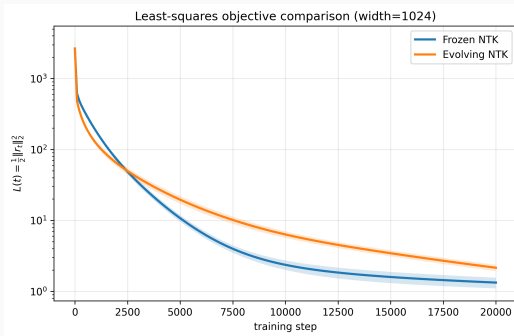
Cross-width residual decay



Effect of kernel evolution depends on width

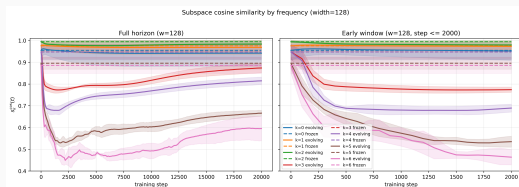


Width $m = 128$: evolving kernel reaches lower training error.

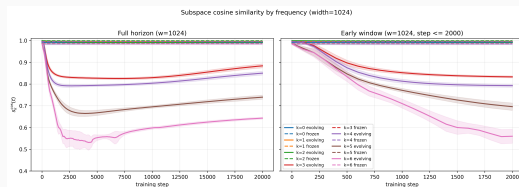


Width $m = 1024$: frozen kernel performs better.

Does kernel evolution change Fourier alignment?



Width $m = 128$: stronger loss of alignment for higher frequencies.

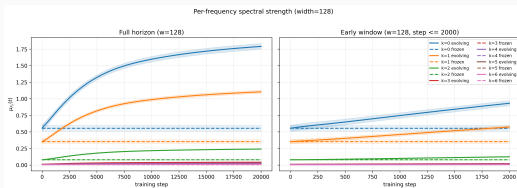


Width $m = 1024$: alignment still degrades, but less severely.

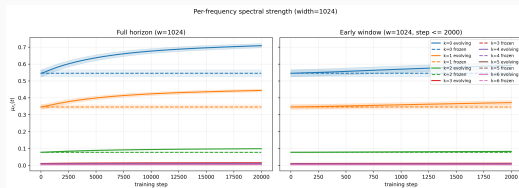
So why can the evolving kernel perform better?

$$\mu_{\mathcal{F}_k}(t) = \frac{1}{d_k} \text{tr} \left(U_k^\top T_t^{(m)} U_k \right)$$

Average kernel strength on the Fourier subspace $\mathcal{F}_k$.



Width $m = 128$: low-frequency spectral strength increases strongly.



Width $m = 1024$: spectral strength changes much less.

Takeaways

1. The infinite-width NTK gives a clean Fourier-diagonal reference:

$$\frac{d}{dt}r_t = -T_\infty r_t, \quad T_\infty \phi_p = \lambda_p \phi_p.$$

2. Finite samples and finite frozen width perturb this picture, but preserve it approximately on $\mathcal{H}_K$.
3. For evolving finite-width networks, the kernel does not simply stay close to the Fourier reference.
4. At small width, kernel evolution can improve optimization by increasing low-frequency spectral strength, even while Fourier alignment worsens.

Main message: spectral bias remains the organizing picture, but finite width changes both the geometry and the effective learning rates.